# E-tiGo — Open Questions Decision Report

Phase 1 Backend — Final Decisions on Outstanding Product & Regulatory Questions

**Prepared by:** Engineering Team | **Date:** 13 October 2026 | **Document version:** 2.0 | **Status:** Decisions Approved & Implemented


**Executive Summary**

During Phase 1 backend development, eight product and regulatory questions remained unresolved. This document records the final decision for each, benchmarked against industry practices from Bolt, Uber, InDrive, and comparable ride-hailing platforms in the Nigerian market.

**All eight decisions have been approved and implemented in the codebase.** Items tagged NEW were implemented in this update cycle (13 Oct 2026). The implementation includes new API endpoints, scheduled jobs, database migrations, and documentation updates.


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## 1. OQ-04 — Emergency Services Dispatch UPDATED

### Question

Which emergency-services dispatch integration should E-tiGo use for its SOS/panic button feature?

### Industry Benchmark

In Nigeria and across African markets, dedicated emergency dispatch APIs (such as RapidSOS) are either unavailable or prohibitively expensive. Bolt Nigeria, Uber Nigeria, and InDrive all route panic events through internal safety teams with structured data, rather than integrating with external dispatch providers. Nigeria's 112 emergency line is the standard public safety number.

### Decision

> **No external dispatch API integration.** The SOS panic button will:
>
> - Send real-time GPS coordinates, trip details, and rider/driver contact information via **WhatsApp Business API** to a monitored internal E-tiGo safety number
> - Simultaneously instruct the mobile client to auto-dial **112** (Nigeria's emergency line) via the `auto_dial_emergency` response field
> - **Notify Admin dashboard in real-time** via WebSocket broadcast on initial trigger (not just escalation) — safety operators see a live incident view
> - Store each incident with encrypted telemetry for audit trail and post-incident review

> **Implementation (13 Oct 2026):**
>
> - `SosIncidentTriggered` event — broadcasts to `admin.sos` WebSocket channel on trigger (instant admin visibility)
> - `SendSosWhatsAppAlertJob` — dispatches GPS + trip + contact details to safety number via WhatsApp Business API
> - `auto_dial_emergency: "112"` — added to SOS trigger API response for mobile client
> - Config: `WHATSAPP_SAFETY_NUMBER` and `WHATSAPP_SOS_TEMPLATE` env vars in `config/services.php`

## 2. OQ-05 — Ride Cancellation Fee Policy

### Question

Should Phase 1 include a cancellation fee, and under what conditions should it apply?

### Decision

> **No cancellation fee in Phase 1.** Riders may cancel freely at any point without penalty. The existing `ride.free_cancellation_window_seconds` configuration is retained for future use when cancellation fees are introduced in a later phase.

### Rationale

Removing friction for early users is more valuable at launch than recovering cancellation costs. The configuration infrastructure is already in place, so adding time-window-based fees later requires minimal engineering effort.

> **Phase 2 recommendation:** Introduce a flat ?300–?500 cancellation fee after the 60-second grace window, with automatic waiver if driver GPS shows no movement toward pickup within 2 minutes of acceptance.

## 3. OQ-06 — Driver Payout Frequency UPDATED

### Question

How often should driver earnings be paid out — on-demand only, or with an automated schedule option?

### Decision

> **Both on-demand and weekly auto-payout.**
>
> - **On-demand**: driver requests via app, Admin approves, processed via Paystack Transfer API
> - **Weekly auto-payout** (opt-in): every Monday at 06:00 WAT, automatic transfer of available balance to driver's bank account — auto-approved for scheduled payouts
> - **Admin-configurable:** payout thresholds adjustable via `AdminWalletSettingsController`

> **Implementation (13 Oct 2026):**
>
> - Migration: `payout_preference` column added to `drivers` table (default: `on_demand`)
> - `PayoutPreference` enum: `on_demand`, `weekly_auto`
> - `PUT /driver/payouts/preference` — driver API to switch preference (requires verified bank account for weekly)
> - `ProcessWeeklyAutoPayoutJob` — scheduled every Monday at 06:00 WAT; processes drivers with `weekly_auto` preference and balance above minimum payout threshold

## 4. OQ-15 — EV Charging Station Data Source UPDATED

### Question

Where does EV station data come from — a third-party API, public dataset, or proprietary entry?

### Decision

> **E-tiGo's own proprietary charging network only.**
>
> - Admin manages stations via CRUD endpoints (including delete)
> - No external data feed integration — only E-tiGo-owned or E-tiGo-approved stations
> - External station owners may apply to join the E-tiGo network via an onboarding form (Phase 2)
>
> **Station data fields:** name, GPS location, stall count, connector types (CCS2/CHAdeMO/Type2/GBT/Tesla), power output (kW), pricing per kWh, operating hours, real-time stall availability.

> **Implementation (13 Oct 2026):**
>
> - Migration: added `connector_types` (JSON), `power_output_kw`, `pricing_per_kwh`, `operating_hours` (JSON) to `ev_charging_stations`
> - `DELETE /admin/ev-stations/{station}` — admin can now delete stations (blocked if active reservations exist)
> - Store/Update form requests updated with validation for new fields

## 5. OQ-17 — Minimum App Version Enforcement NEW

### Question

What are the minimum supported iOS and Android versions, and how is version enforcement handled?

### Decision

> **Minimum version set to launch version (1.0.0 for both iOS and Android).**
>
> - `min_app_version_ios` / `min_app_version_android` stored in AppSettings — Admin-adjustable at runtime
> - Returned via the `GET /api/v1/config` public endpoint
> - Mobile app checks on every launch and shows a "please update" screen when the installed version is below minimum
> - `force_update` flag available for critical updates that block app usage entirely

> **Implementation (13 Oct 2026):**
>
> - `ConfigController` — new `GET /api/v1/config` public endpoint (no auth required)
> - Returns: `min_app_version.ios`, `min_app_version.android`, `force_update`, `emergency_number`
> - Values read from AppSettings with sensible defaults (1.0.0, false)

## 6. OQ-27 — Refund Policy

### Question

Should refunds go to the passenger's wallet as credit, or back to the original payment method (card)?

## Decision

> **Wallet-credit-only refunds for V1.**
>
> - All refunds (full and partial) issued as instant wallet credits via `WalletRefundService`
> - Card refund available as an **admin-only escalation path** — used exclusively by support admins resolving disputes via `AdminDisputeController::resolve()`. Not accessible to end users
> - Card refund capability may be expanded in Phase 2 if required by CBN consumer protection regulations

## Rationale

Wallet credits are instant (better UX), cheaper to process (no Paystack refund fees), and reduce chargeback exposure. The admin-escalation path through dispute resolution ensures high-value edge cases can be handled via card refund when warranted.

## 7. OQ-32 — Tips in Wallet Flow

### Question

Should riders be able to tip drivers by debiting their E-tiGo wallet balance?

### Decision

> **No wallet-debit tips in V1.**
>
> - **Card tips:** supported — charges the rider's card via Paystack
> - **Cash tips:** handed directly to the driver; logged as `tip_amount` on the Payment record
> - Wallet-debit tips deferred to Phase 2 — only if rider demand is demonstrated and regulatory clarity obtained

### Rationale

Deducting from a closed-loop wallet for tips creates accounting ambiguity (payment vs. transfer) and potential regulatory issues under CBN guidelines. Card-only tips are the proven, compliant approach used by all major platforms.

## 8. OQ-33 — Regulatory Compliance for Closed-Loop Wallet

### Question

Does E-tiGo need a separate PSP or MMO license to hold customer funds in its closed-loop wallet?

### Decision

> **E-tiGo operates under Paystack's CBN-licensed PSP infrastructure.**
>
> - The wallet system uses `PaystackWalletGateway` — all customer funds are held in Paystack's regulated trust accounts
> - E-tiGo functions as a sub-wallet under Paystack's license
> - Additional safeguards: ?500,000 max wallet balance, double-entry bookkeeping ledger, full audit trail

> **Action required:** Execute a formal partnership agreement with Paystack confirming the sub-wallet arrangement and regulatory coverage before public launch.

## 9. Decision Summary Matrix

| # | Question | Decision | Code Status | Notes |
|---|---|---|---|---|
| OQ-04 | Emergency dispatch | WhatsApp + 112 auto-dial + Admin real-time alerts | **Implemented** | UPDATED Broadcast + WhatsApp + 112 |
| OQ-05 | Cancellation fee | No cancellation fee in Phase 1 | **Aligned** | Config retained for Phase 2 |
| OQ-06 | Payout frequency | On-demand + weekly auto-payout (opt-in) | **Implemented** | NEW Job + preference + migration |
| OQ-15 | EV station data | Proprietary network, full CRUD, detailed fields | **Implemented** | UPDATED New columns + delete |
| OQ-17 | Min app versions | 1.0.0 at launch, server-side enforcement | **Implemented** | NEW /api/v1/config endpoint |
| OQ-27 | Refund policy | Wallet-credit default, card refund admin-only escalation | **Aligned** | Both paths exist, correctly scoped |
| OQ-32 | Tips in wallet | Card tips only, no wallet-debit | **Aligned** | No wallet tip path exists |
| OQ-33 | Wallet regulatory | Operate under Paystack's PSP license | **Aligned** | Paystack is sole wallet gateway |

## 10. Implementation Status

The following code changes were made on 13 October 2026 to align all decisions with the codebase:

| Change | Files Modified/Created | Details |
|---|---|---|
| SOS real-time admin broadcast | `SosIncidentTriggered` (new event)<br>`SosService.php` | Broadcasts to `admin.sos` on trigger, not just escalation |
| SOS WhatsApp safety alert | `SendSosWhatsAppAlertJob` (new)<br>`config/services.php` | GPS + trip details to safety number via WhatsApp Business API |
| SOS 112 auto-dial signal | `SosController.php` | `auto_dial_emergency: "112"` in trigger response |
| Weekly auto-payout | `ProcessWeeklyAutoPayoutJob` (new)<br>`routes/console.php` | Scheduled Monday 06:00 WAT, auto-approves for opted-in drivers |
| Driver payout preference | Migration (new)<br>`Driver.php`<br>`PayoutController.php`<br>`routes/api.php` | `PUT /driver/payouts/preference` + `PayoutPreference` enum |
| EV station detail fields | Migration (new)<br>`EvChargingStation.php`<br>Form requests | connector_types, power_output_kw, pricing_per_kwh, operating_hours |
| EV station delete | `AdminEvStationController.php`<br>`routes/api.php` | `DELETE /admin/ev-stations/{station}` |
| App config endpoint | `ConfigController` (new)<br>`routes/api.php` | `GET /api/v1/config` — min version + force update flag |

## 11. Action Items & Next Steps

| # | Priority | Action | Owner | Timeline |
|---|---|---|---|---|
| 1 | HIGH | Execute formal sub-wallet partnership agreement with Paystack confirming regulatory coverage for closed-loop wallet | Legal / BD | Before public launch |
| 2 | HIGH | Set up `WHATSAPP_SAFETY_NUMBER` and create the `sos_emergency_alert` WhatsApp template in Meta Business Manager for production | Ops / Engineering | Before public launch |
| 3 | MEDIUM | Coordinate with mobile team on: (a) 112 auto-dial implementation using `auto_dial_emergency` response field, (b) force-update gate using `/api/v1/config` endpoint | Mobile Team | Sprint 1 post-approval |
| 4 | LOW | Design Phase 2 "apply to charge" onboarding form for external EV station partners joining E-tiGo's network | Product | Phase 2 planning |
| 5 | LOW | Plan Phase 2 cancellation fee implementation: ?300–?500 flat fee after 60s grace window, with driver no-movement waiver | Product | Phase 2 planning |